

Use-Case Flow: Compute Debt Settlement

Alternate Flow — Partial Settlement Confirmation

Trip Expense Simplification Engine — Problem Statement #37

Context

This alternate flow extends the Main Success Scenario of the "Compute Debt Settlement" use case (see Use-Case Flow: Compute Debt Settlement). It covers the case where a Group Traveler confirms only some of the proposed settlement edges instead of all of them in one sitting, and directly realizes requirement FR-001 (the system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions), since a partial confirmation must not invalidate the previously computed minimal edge set.

Trigger

At step 6 of the Main Success Scenario, the Group Traveler confirms one or more — but not all — of the displayed settlement edges as completed.

Alternate Flow: 6a. Partial Settlement Confirmation

6a1. Group Traveler selects a subset of the displayed settlement edges and marks them as paid, leaving one or more edges unconfirmed.

6a2. System marks only the selected edges as paid and recalculates the remaining net balances for the unconfirmed edges.

6a3. System displays the trip status as “Partially Settled” and highlights which edges are still outstanding.

6a4. Group Traveler may leave the screen and return later; the unconfirmed edges remain visible and unchanged until acted on.

6a5. Use case resumes at step 5 (Group Traveler reviews the proposed settlement plan) the next time the Group Traveler opens “Settle Up” for the trip, until every edge is confirmed and step 8 is reached.

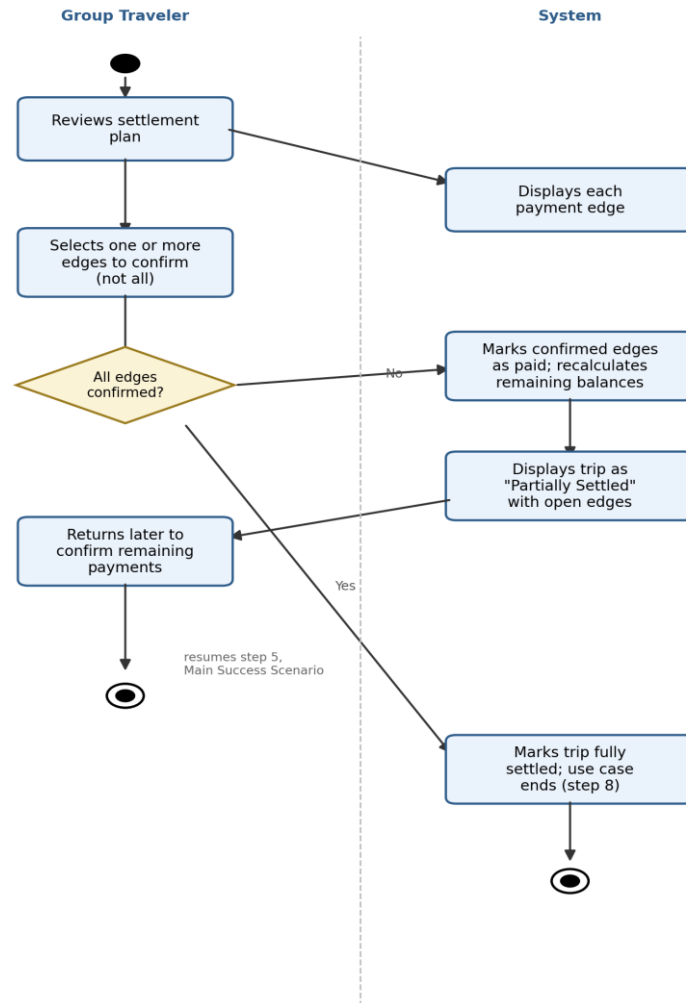
Postconditions

- Success: confirmed edges are marked paid and excluded from future settlement recalculation; unconfirmed edges remain open.
- The minimal-edge property from FR-001 still holds for the remaining unpaid balances, since recalculation is scoped to the outstanding travelers only.

Activity Diagram

The diagram below models this alternate flow as a two-lane activity diagram (Group Traveler / System).

Activity Diagram — Alternate Flow: Partial Settlement Confirmation (FR-001)



Traceability

Requirement	FR-001 — optimal debt settlement graph via min-flow algorithm
Base Use Case	Compute Debt Settlement
Extends Step	Step 6 of the Main Success Scenario
Actors	Group Traveler (primary), System